

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 23560

LM INFERENCE FROM SCIENTIFIC DATA

SEMESTER 1 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

ANY CALCULATOR

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. (i) Write down the definitions of the mean and variance of a continuous, real-valued random variable x with a probability density function (PDF) of $P(x)$. You should give your answers in terms of integrals. **[3]**

A real-valued random variable t follows a Student's t -distribution with $\nu > 2$ degrees of freedom. The PDF for this distribution is given by

$$f_{\nu}(t) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-\left(\frac{\nu+1}{2}\right)}.$$

- (ii) Find the mean of the Student's t -distribution, $\langle t \rangle$. **[2]**

In the next part of the question you may use the following property of the Gamma function: $\Gamma(\alpha + 1)/\Gamma(\alpha) = \alpha$. You may also use the following standard integral,

$$\int_0^{\infty} dx \frac{x^2}{(1 + x^2)^{\alpha}} = \frac{\sqrt{\pi}}{4} \frac{\Gamma(\alpha - \frac{3}{2})}{\Gamma(\alpha)}.$$

- (iii) Show that the variance of the Student's t -distribution with $\nu > 2$ degrees of freedom is given by $\text{Var}(t) = \nu/(\nu - 2)$. **[18]**
- (iv) For a particular value of ν , e.g. $\nu = 3$, sketch the Student's t -distribution. On the same set of axes also sketch the normal distribution with the same mean and variance. **[7]**

2. A biased coin has a probability α of giving a heads when it is flipped. The coin is flipped 5 times and $n = 2$ heads are obtained.

(i) Write down an expression for the likelihood $P(n|\alpha)$ for these 5 flips. **[3]**

In the remainder of the question you should use a flat, uniform prior on the unknown probability parameter α in the range $0 < \alpha < 1$.

(ii) Using Bayes' theorem, obtain an expression the posterior $P(\alpha|n)$. You should make sure your posterior distribution is correctly normalised. **[5]**

(iii) Sketch the posterior distribution $P(\alpha|n)$ from part (ii) as a function of α . Mark clearly on your plot the location of the peak of the distribution. **[3]**

Now the coin is flipped many more times until the total number of flips performed is N . The total number heads obtained is n .

(iv) Write down a new expression for the posterior $P(\alpha|n)$ for all N flips, this time you do NOT need to normalise the distribution. Find the most likely posterior value $\hat{\alpha}$ for the coin probability parameter. **[3]**

(v) Write down a Taylor series for $\log P(\alpha|n)$ expanding about the point $\alpha = \hat{\alpha}$ and keeping terms up to and including those of order $(\alpha - \hat{\alpha})^2$. **[5]**

(vi) Using the expression from part (v), or otherwise, estimate the uncertainty on the parameter α . State the conditions under which your estimate is likely to be a good approximation to the true standard deviation. **[6]**

(vii) Estimate the number times the coin must be flipped in order to measure the parameter α to a fraction uncertainty of 0.1%? You need only give your answer to the nearest order of magnitude. **[5]**

3. The capital letters A , B and C represent events or propositions. The symbol $\bar{A}$ denotes the event “not A ” and $P(A)$ denotes the probability of event A occurring. The notation $P(A|B)$ means the “probability of A given B ”.

Draw Venn diagrams to illustrate the following rules of probability.

- (i) The sum rule $P(A) + P(\bar{A}) = 1$. **[2]**
- (ii) The extended sum rule $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. **[3]**
- (iii) The product rule $P(A \text{ and } B) = P(A)P(B|A)$. **[4]**

The following equation is known as Bayes’ theorem,

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)}.$$

- (iv) Using the product rule, prove Bayes’ theorem **[3]**

A random variable $x > 0$ follows a gamma distribution with constant shape parameters $\alpha > 0$ and $\beta > 0$ if it has a probability density function

$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} \exp(-\beta x)}{\Gamma(\alpha)} \quad \text{for } x > 0.$$

You may assume that this distribution is correctly normalised such that

$$\int_0^\infty dx f(x; \alpha, \beta) = 1.$$

Decay events in a large radioactive sample occur independently and at a constant average rate r per unit time. In a time T a number n events are observed.

- (v) What standard probability distribution is appropriate to use for the likelihood in this situation? Write down the standard expression for the the likelihood $P(n|r)$. **[4]**

In the remainder of the question you should use a gamma distribution with parameters α and β as a prior distribution for the rate parameter, $f(r; \alpha, \beta)$.

- (vi) Use Bayes’ theorem to show that the posterior distribution on the rate parameter r is also a gamma distribution. Write down the posterior PDF on the parameter r making sure the distribution is correctly normalised. **[14]**

Formula Sheet

Binomial distribution.

$$P(r; n, p) = p^r (1 - p)^{n-r} C_r \quad (1)$$

Poisson distribution.

$$P(n; \lambda) = \frac{\lambda^n \exp(-\lambda)}{n!} \quad (2)$$

Normal distribution. Also known as the Gaussian distribution.

$$P(x; \mu, \sigma^2) = \frac{\exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \quad (3)$$

Bayes' theorem.

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)} \quad (4)$$

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[N.B. 1 mole $\equiv$ 1 gram-molecule]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2})$

ANY CALCULATOR

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.